Matthew Topham

AI Systems & Full-Stack Developer

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Computer science student at CU Boulder building AI tools for spatial creation, wearable context, and interactive storytelling. Work spans TypeScript interfaces, Python services, model integration, persistent state, and automated validation.

Selected projects

School of the Ancients + Matrix · Spatial AI learning

Creator · 2024–Present · Quest 3 / WebXR

- Built an embodied learning experience where a learner's question can lead to a Blender-authored exhibit, loaded into a persistent Three.js/WebXR world for room-scale exploration.
- Connected teacher conversations, avatar voices, selected-part context and optional GPT live voice / Decisions API to asynchronous creation and world restoration.
- Developed the underlying Matrix voice/text creation interface, procedural and GLB asset workflows, spatial interaction, saved worlds and release validation.

Quest demo · Report and slides · Source · October rc.7 checkpoint

Manfred · Wearable AI memory and context

Creator · 2026 · Prototype

- Built an Android/Flutter companion and Python services for wearable audio ingestion, transcription, searchable context, and durable upload buffering.
- Connected E09 smart glasses to ChatGPT Live through phone automation; voice works, while photo capture and delivery remain under refinement.

Open-source experience

Marinara Engine + Agents · Pasta Devs

Open-Source Developer · 2026–Present

- Contributed 100 merged pull requests across Engine (68) and Agents (32), spanning React/TypeScript UI, Node.js services, persistent world state, and regression tests.
- Built hierarchical World Maps and versioned host APIs for downloadable agents, including persistence, transaction handling, and recovery paths.
- Implemented image-aware storyboard planning that grounds animation in generated keyframes; packaged live-tested MiniMax H3 prompts as reusable Storyboard choices.

Selected work: World Maps · host APIs · animation planning · H3 prompts

Additional work

Demerzel: Built conversation-to-wiki ingestion, daily AI briefs, service health checks, and GPU workload controls for a local AMD/NVIDIA compute fleet.

Orbital Guardian: Integrated Unity space-scene experiments into a PC VR station-defense prototype with handjet locomotion, moving handholds, ship piloting and recovery controls.

Technical skills

Software: TypeScript, JavaScript, Python, SQL, React, Node.js, Flutter, REST, WebSockets.
Spatial / AI / infrastructure: Three.js, WebXR, Blender, local inference, ComfyUI, Linux, Docker, GitHub Actions, Playwright, regression testing.

Education & certification

University of Colorado Boulder — B.A. Computer Science, in progress
Front Range Community College — A.S. Computer Science, 2025
NVIDIA-Certified Associate — AI Infrastructure & Operations, August 2025